ESABCC Indicator Database — Full Fact-Check

Raw data & derivation audit of all 97 progress indicators

Generated 2026-06-03. Scope: every numeric value in *src/data/esabcc-indicators.ts* (97 indicators) plus the 19 percent-unit *ecno-indicators.ts* entries used for cross-checking. Three axes were verified: **(1) raw-data integrity** — committed values vs the report's published underlying-data workbook (extract.json); **(2) derivation correctness** — the 45 reverse-engineered chapter derivations recomputed from raw workbook rows; **(3) cross-source consistency & weird-calculation hunt** — chapter working files vs consolidated data, unit/sign/scale conventions, and the new completeness-pass indicators (I2, A2, A4, A5, A6, B3, B4, L2, I7a/b/c). Verification used the chapter workbooks, openpyxl recomputation and seven parallel auditing agents (one per chapter).

Severity	Count	Meaning
HIGH	5	Wrong as displayed / 100x scale error — fix before relying on the figure
MEDIUM	13	Label/scale/completeness defect — value faithful but presentation/coverage off
LOW	16	Cosmetic / housekeeping / minor rounding
INFO	128	Verified clean, or context note (no action)

Headline conclusion

The underlying numbers are sound. Every report-derived value in the database reproduces the ESABCC report's own published underlying data to 4-significant-figure rounding, and all 45 derivations recompute their published figures from raw rows to within 0.06% (most exactly). No transcription error, sign flip, frozen series or year-offset survived the check.

The one material defect is a presentation-scale inconsistency, not a wrong value. 13 ESABCC indicators carrying unit '%' store fractions (0–1) while the 19 ECNO indicators and 2 ESABCC ones (A3-NUE, F4) carrying the same unit store percent-numbers. Because the renderer prints *value + unit* with no ×100, the 13 fraction-stored series display 100× too small, and several sit on the opposite scale to the very ECNO indicator they are flagged *duplicateOf* (e.g. fossil power share 0.52 vs 42.0). This is the headline fix.

1 · Cross-cutting findings

HIGH 100x percent-encoding inconsistency across the indicator database

The display path (IndicatorDetail.tsx:79 and IndicatorChart.tsx:90) renders `value.toLocaleString() + ' ' + unit` with NO x100 scaling. The 19 ECNO indicators with unit '%' store the PERCENT NUMBER (e.g. fossil-power-share = 42.0, ev-share-new-cars = 1.9, building-renovation-rate = 1.0). But 13 ESABCC indicators with unit '%' store FRACTIONS (0-1): E2 fossil (0.5183), E2 RES (0.1426), E5 (0.2123), I3 (0.108), I6 (0.3171), T3a (0.8128), T5a (0.0001), T6a (0.9769), B5a (0.5661), B5b (0.4698), F2-share (0.0008), B3-residential (0.010), B3-commercial (0.006). These render 100x too small (e.g. E2 fossil shows '0.518 %' instead of 51.8%; B3 shows '0.01 %' instead of 1.0%). Within unit '%', only A3-NUE (42-52) and F4 (11.9-14.3) use the correct percent-number convention.

Concept	ESABCC (stored)	ECNO duplicateOf	Gap
Fossil share of electricity	E2 (fossil) = 0.5183 (2005)	fossil-power-share = 42.0	~100x
Electricity share of final/end-use energy	E5 = 0.2123	end-use-electrification = 22.1	~100x
Electricity share of industrial energy	I6 = 0.3171	industry-electrification = 33.0	~100x
Road share of passenger transport	T3a = 0.8128	road-passenger-share = 82.7	~100x
ZEV/BEV share of new cars	T5a = 0.0001..0.146	ev-share-new-cars = 1.9..12.1	~100x
Building renovation rate	B3-residential = 0.010	building-renovation-rate = 1.0	~100x

Recommendation: Pick one convention (percent-number, matching the ECNO majority and the no-scaling renderer) and multiply the 13 fraction-stored ESABCC '%' series by 100; OR add a single display-time x100 for unit '%'/'% of GDP' (and fix the two already-percent-number ones to match). Either way, duplicateOf pairs must end up on the same scale. The underlying VALUES are faithful to the report; only the stored scale/label is inconsistent.

MEDIUM B4 series carry unit '% of 2005' but store an index ~1.0

All five B4 series (dwellings, floor-area, surface-residential, population, surface-tertiary) store an index normalised to 2005 = 1.0 (range 0.996..1.131) but declare unit '% of 2005'. Rendered raw this reads e.g. '1.131 % of 2005', which is meaningless. Either the values should be x100 (so 113.1 % of 2005) or the unit relabelled to 'index (2005=1.0)'.

MEDIUM I2 cement trade balance missing from the database

extract.json Figure 24 position 7 ('cement trade balance') is a byte-identical CORRUPT copy of the steel trade-balance series; the build recipe correctly omitted it. But the genuine cement trade balance does exist in the chapter workbook (I2 sheet rows 18-20: 2005=-2.316 ... 2021=+3.492 Mt) and was never imported. Result: the DB has steel and chemicals trade balances but no cement one, while the cement production/use descriptions still say 'Production, use and trade balance together show...'. Either import the real cement trade series from the workbook or amend the two cement descriptions.

LOW Stale duplicate Ch8 workbook committed to the repo

Two Ch8 workbooks are committed: 'Ch8. Agriculture - final indicators and graphs.xlsx' (canonical) and 'Ch8. Agriculture - final indicators and graphs-laptop9883.xlsx' (a stale laptop copy). Diffs are confined to non-published auxiliary rows (the laptop file lacks the published NUE row and has an offset A1 scenario block). No published value is affected, but the divergent duplicate is a source-control risk; recommend deleting the laptop copy.

MEDIUM E4b/c label says 'onshore + offshore' but data is onshore-only

E4b/c is named/described as wind additions 'onshore + offshore (all axes)' but the published Historic series and derivation use ONSHORE ONLY (workbook row81). Total wind (row80) would differ materially. The value is faithful to extract.json, but the name overstates offshore inclusion.

LOW Uneven afterReport (live-source) coverage; E4a/E4b skip 2022

E4a and E4b afterReport tails skip 2022, jumping from the report's 2021 directly to 2023/2024 (a one-year hole in the live-source data). Several other indicators also have uneven afterReport coverage (some end 2023, some 2024).

2 · Ch3

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
O1	esabcc-o1-ghg-total	✓	✓	0	0	clean
O2 (PEC)	esabcc-o2-pec	✓	✓	0.4545	0.00284	clean
O2 (FEC)	esabcc-o2-fec	✓	✓	0.4591	0.00403	clean
O3	esabcc-o3-gross-inland	✓	✓	0.3265	0.002	clean

Notes: All four Ch3/Overall indicators (O1, O2-PEC, O2-FEC, O3) pass cleanly. (1) Raw-data integrity: ts_points 2005-2021/2022 reproduce the extract.json published series exactly; the only non-zero raw delta is O1 2005 = 0.0456 Mt, attributable to 1-decimal display rounding in the .ts. (2) Derivation correctness: the ch3.py recompute matches extract.json overview to <=0.46 TWh (O2) / <=0.33 TWh (O3) / 0.0 (O1); these sub-TWh gaps are entirely due to extract.json storing O2/O3 as integers while the derivation/workbook carry full precision (relative deltas <=0.004%). (3) Cross-source consistency: the detailed Ch3 working workbook ('O2. Primary & final energy use' rows 6/7, 'O3. gross inland consumption' row3, 'O1 & Fig 2. GHG emissions' row3) computes the same figure series as the consolidated extract.json, with cached cell values matching to full float precision - the two workbooks agree. (4) Unit conversions all correct: Mtoe->TWh = 11.63 (verified in 'conversion factor' sheet E6), ktoe->TWh = 0.01163 (E5); EED 763 Mtoe benchmark -> 8874 TWh independently confirms factor. (5) afterReport 2022-2024 points (O2/O3) are plausible energy-crisis-era declines continuing the historical downtrend, correct magnitude, O3>PEC>FEC ordering preserved. One methodological note worth recording (not an error): O1 'Total (ECL scope)' scales international maritime by V68=0.6375, a year-derived EU-ETS coverage ratio (S66/S67) rather than a fixed constant; aviation is counted in full. No errors, no frozen constants, no sign/offset/flat-series anomalies found.

3 · Ch4

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
E1	esabcc-e1-energy-supply-ghg	✓	✓	0.4851	0.0572	clean
E2 (fossil)	esabcc-e2-fossil-power-share	✓	✓	0	0	medium
E2 (RES)	esabcc-e2-res-noBio-power-share	✓	✓	0	0	medium
E3	esabcc-e3-grid-co2-intensity	✓	✓	0	0	clean
E4a	esabcc-e4a-solar-pv-add	✓	✓	1.377e-14	3.61e-13	clean
E4b/c	esabcc-e4b-wind-add	✓	✓	1.243e-14	1.76e-13	medium
E5	esabcc-e5-electrification	✓	✓	0	0	medium
E6	esabcc-e6-energy-ch4	✓	✓	0	0	clean

Notes: All 8 Ch4 energy indicators: raw_data_ok and derivation_ok TRUE. ts non-afterReport points match extract.json within 4-sig-fig rounding (worst raw rel 0.043% on E6 2005). Derivations reproduce extract overview: E2/E3/E5/E6 exact (0 delta); E4a/E4b exact to ~1e-14; E1 worst 0.485 abs (0.057%) at 2016 purely from publishing the Total as an integer (components sum exactly to 1168.5149). WEIRD-CALC AUDIT: (E1,E6) manual CRF sums internally consistent, components sum to published total every year, no frozen/stale cached external-link values; E1 2022 published-total override correct. (E2) share ratios correct, denominator=total gross generation, fossil+nuclear+RESnonbio+bio sum to 1.0, biomass correctly excluded from RES-nonbio. (E3) passthrough, unit & magnitude (228-386 g/kWh) correct. (E4a/E4b) annual additions = cum(t)-cum(t-1) verified, first year handled, NO negative spurious additions; workbook cumulative rows mislabelled 'GW' but are MW and /1000 conversion is correct. (E5) electricity/FEC ratio correct. KEY FINDINGS: [MEDIUM] E2(fossil), E2(RES), E5 declare unit '%' but store fractions 0-1 (systematic, consistent with extract — display-convention risk, not a data error). [MEDIUM] E4b/c is labelled 'wind additions, onshore+offshore' but published Historic series & derivation are ONSHORE ONLY (row81), excluding offshore. [LOW] E4a & E4b afterReport series skip 2022 (jump 2021->2023), leaving a one-year gap in the live data. No sign errors, no hardcoded constants beyond legitimate /1000 and GWP factors, no frozen series, no year offsets, no rounding anomalies beyond documented publish-rounding.

Flagged items:

MEDIUM E2 (fossil) — [unit-value-mismatch] unit field is '%' but ts values are fractions 0-1 (2005=0.5183, 2022=0.3876), not 0-100. Consistent with extract.json (also fraction + '%'), so it is a systematic display convention rather than a per-point error, but a literal value+unit render would show '0.52 %' instead of '51.8 %'.
MEDIUM E2 (RES) — [unit-value-mismatch] unit '%' but ts values are fractions 0-1 (2005=0.1426, 2022=0.3376). Same systematic convention as E2 fossil; consistent with extract.json.
MEDIUM E4b/c — [scope-vs-label] Indicator labelled E4b/c 'Annual wind capacity additions (all axes), onshore + offshore', but the published Historic series (extract Figure 17) and derivation use ONSHORE ONLY (workbook row81), NOT total wind (row80). Verified: extract matches onshore exactly every year (e.g. 2011 extract=13.616 = onshore 13.616, total wind=8.470); total wind would differ materially. Faithful to extract, but the name/description overstating offshore inclusion is misleading.
MEDIUM E5 — [unit-value-mismatch] unit '%' but ts values are fractions 0-1 (2005=0.2123, 2021=0.2275), not 0-100. Same systematic convention as E2; consistent with extract.json.

4 · Ch5

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
I1	esabcc-i1-industry-ghg	✓	✓	0.04905	0.0064	clean
I3	esabcc-i3-circular-mat-use	✓	✓	0	0	clean
I4 (steel)	esabcc-i4-steel-ghg-intensity	✓	✓	0.0003753	0.037	clean
I4 (cement)	esabcc-i4-cement-ghg-intensity	✓	✓	4.951e-05	0.0073	clean
I4 (chemicals)	esabcc-i4-chemicals-ghg-intensity	✓	✓	4.826e-05	0.0061	clean
I5	esabcc-i5-industry-fec	✓	✓	0.04697	0.0017	clean
I6	esabcc-i6-industry-electrification	✓	✓	4.984e-05	0.015	clean
I2 (steel, production)	esabcc-i2-steel-production	✓	—	0.04444		clean
I2 (steel, use)	esabcc-i2-steel-use	✓	—	0.044		clean
I2 (steel, trade)	esabcc-i2-steel-trade-balance	✓	—	0.004777		clean
I2 (cement, production)	esabcc-i2-cement-production	✓	—	0.049		clean
I2 (cement, use)	esabcc-i2-cement-use	✓	—	0.047		medium
I2 (chemicals, production)	esabcc-i2-chemicals-production	✓	—	0.003702		clean
I2 (chemicals, use)	esabcc-i2-chemicals-use	✓	—	0.004954		clean
I2 (chemicals, trade)	esabcc-i2-chemicals-trade-balance	✓	—	0.0004852		clean
I7a	esabcc-i7a-steel-projects	✓	—	0		clean
I7b	esabcc-i7b-cement-projects	✓	—	0		clean
I7c	esabcc-i7c-chemicals-projects	✓	—	0		clean

Notes: Ch5/Industry audit: all 16 indicators verified. DERIVED (I1,I3,I4 steel/cement/chemicals,I5,I6): ch5.py recompute reproduces extract.json published series with worst_abs=0.0 (exact) for every series; ts_points vs extract deltas are pure rounding (<=0.049, all <0.01%). Unit conversions verified: I1 kt->Mt (/1000), I5 Mtoe->TWh (x11.63, conversion-factor sheet E6 confirmed), I3/I6 stored as fractions with unit '%'. afterReport points (I1 2023/24, I3 2022-24, I5 2022/23, I6 2022/23) all plausible vs trend. NEW I2 (8 series): ts match extract.json Figure 24 at correct seriesPos 0,1,2,5,6,10,11,12 exactly (rounding only). NEW I7a/b/c counts (48/62/171) all reconcile with workbook breakdowns and .ts descriptions arithmetically. CEMENT-TRADE-OMISSION VERDICT: CONFIRMED CORRECT to omit extract.json Figure 24 pos7. pos7 is a byte-identical copy of steel trade pos2 (same values incl. 2009=0.14275, same 2008-start while cement prod/use start 2005). The workbook (I2 sheet rows18-20) contains the REAL, DIFFERENT cement trade balance (D20=-2.316 ... T20=+3.492, 2005-2021); extract.json pos7 is a stray steel copy. Consequence: the published DB has NO cement trade balance series at all (8 I2 entries, no cement-trade) -- a completeness GAP (medium), not a data error; recipe correctly avoided importing the corrupt pos7. Minor doc nit: cement production/use descriptions still say 'production, use and trade balance together' though trade is absent. I2-STEEL-2009 VERDICT: 0.1428 Mt is GENUINE (workbook H9 = exports 26.6728 - imports 26.5301 = 0.14275). 2009 crisis year both flows collapsed and balanced; the value sitting between -19.02(2008) and -2.915(2010) is real, not a sign/transcription error. No errors found.

Flagged items:

MEDIUM **I2 (cement, use)** — [completeness-gap] DB omits cement trade balance entirely: extract.json Figure 24 pos7 is a stray byte-identical COPY of steel trade (pos2), starting 2008 while cement prod/use start 2005. The genuine cement trade balance exists in the workbook (I2 sheet rows18-20: 2005=-2.316 ... 2021=+3.492 Mt) but was never imported. Recipe's omission of corrupt pos7 is correct; result is a missing-series gap, not a wrong value.

5 · Ch6

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
T1	esabcc-t1-transport-ghg	✓	✓	0	0	clean
T2a	esabcc-t2a-passenger-demand	✓	✓	0	0	clean
T2b	esabcc-t2b-freight-demand	✓	✓	0	0.0019	clean
T3a	esabcc-t3a-road-share-passenger	✓	✓	2.22e-16	2.7e-14	clean
T3b	esabcc-t3b-air-passenger	✓	✓	0	0	clean
T4	esabcc-t4-car-co2-intensity	✓	✓	0	0	clean
T5a	esabcc-t5a-zev-share-newcars	✓	✓	0	0	clean
T5b	esabcc-t5b-zev-lorries-stock	✓	✓	0	0	clean
T6a	esabcc-t6a-fossil-transport-share	✓	✓	0	0	clean
T6b	esabcc-t6b-foodcrop-biofuels	✓	✓	0	0	clean

Notes: All 10 Chapter 6 / Transport indicators (T1, T2a, T2b, T3a, T3b, T4, T5a, T5b, T6a, T6b) are clean. Raw-data integrity (ts non-afterReport points vs extract.json) matches at 4-sig-fig tolerance for every series. Derivation recompute (ch6.build + lib.recompute_layout) reproduces extract.json to within float noise (worst abs delta across all indicators = 2.2e-16 on T3a; all others exactly

0.0). Cross-source consistency holds because the derivation reads the working workbook directly and its output equals extract.json. NO HIGH or MEDIUM severity issues found. Two LOW cosmetic items: (1) T2b 2006 published 2193.5 vs exact 2193.4576 straddles a 4-sig rounding boundary (2194 vs 2193) with true delta 0.0425/0.0019% -- benign. (2) T6b working-file year-header row is mislabeled (D2=2011) while data maps to 2005-2015; derivation hard-maps correctly so published values are right. Unit/magnitude checks all pass: T2a Gpkm ~5000-6000, T2b Gtkm ~2000-2300, T3b Gpkm ~360-585, T5b is an absolute vehicle COUNT not a %, T6b TWh via verified 11.63 Mtoe->TWh factor. Shares T3a/T5a/T6a are stored as 0-1 fractions with unit '%' -- this is the consistent database-wide convention (display x100), not a fraction-vs-percent error. T4 has only 3 historic points because EEA WLTP measurement began in 2020 (replacing NEDC). afterReport extensions (T1 2023-24, T4 2023, T5a 2023-24) are all trend-plausible; T5a 2024=0.136<2023=0.146 reflects the real 2024 EU BEV-sales dip. No sign errors, no g-vs-kg mismatches, no frozen series, no year offsets in published data.

6 · Ch7

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
B1	esabcc-b1-buildings-ghg	✓	✓	0.04783	0.00721	clean
B2	esabcc-b2-buildings-fec	✓	✓	0.04975	0.00112	clean
B5a	esabcc-b5a-residential-fossil-share	✓	✓	4.642e-05	0.00996	clean
B5b	esabcc-b5b-tertiary-fossil-share	✓	✓	4.901e-05	0.0129	clean
B6	esabcc-b6-heat-pump-stock	✓	✓	0	0	clean
B4 (dwellings)	esabcc-b4-dwellings	✓	—	0.0004794	0.0455	medium
B4 (floor area)	esabcc-b4-floor-area	✓	—	0.0004579	0.0438	medium
B4 (surface residential)	esabcc-b4-surface-residential	✓	—	0.0004548	0.0447	medium
B4 (population)	esabcc-b4-population	✓	—	0.0004804	0.0468	medium
B4 (surface tertiary)	esabcc-b4-surface-tertiary	✓	—	0.0004635	0.0458	medium
B3 (residential)	esabcc-b3-residential-renovation-rate	✓	—	0	0	high
B3 (commercial)	esabcc-b3-commercial-renovation-rate	✓	—	0	0	high

Notes: All five DERIVED indicators (B1,B2,B5a,B5b,B6) are CLEAN: TS vs extract.json deltas are pure DB rounding (B1 0.048 Mt/0.007%, B2 0.050 TWh/0.0011%, B5a 4.6e-5/0.010%, B5b 4.9e-5/0.013%, B6 exact 0); derivation recompute vs extract.json matches to machine epsilon (B1/B2 ~1e-13, B5a/B5b/B6 exactly 0). B5a/B5b fossil shares are valid 0-1 fractions (42-57% residential, 32-47% tertiary). B6 unit/magnitude correct (1.1-19.79 million units, REPowerEU 60M target). NEW completeness pass: B4 (Figure 50) all 5 seriesPos positions map correctly to extract labels (0=dwellings,1=floor area,2=surface residential,3=population,4=surface tertiary) and match within 3-dp rounding (<5e-4 abs, <0.047%). B3 (Figure 49) values match the workbook 'B3. (Retrofit)' sheet exactly (residential D10=0.01, commercial H20=0.006; targets E11=0.018, I11=0.011) -- note extract.json Figure 49 has EMPTY data arrays so verification was done against the workbook. B4 INDEX-SCALE VERDICT: values are stored as ratios with 2005 base = 1.0 (range ~0.996-1.131), NOT ~100. Unit label '% of 2005' is therefore a scale ambiguity (MEDIUM): the values are 100x smaller than the label implies; they render correctly only if the frontend multiplies by 100. B3 FRACTION-VS-PERCENT VERDICT: values ARE stored as fractions (0.010=1.0%, 0.006=0.6%) with unit '%' -- this is the chapter's consistent house style (B5a/B5b fossil shares stored identically as 0.x fractions with unit '%'), so a frontend that applies x100 for '%'-unit indicators renders B3 correctly as 1.0%/0.6%. However, a naive value+'%' renderer would display '0.01%'/'0.006%' (100x too small) -- flagged HIGH because B3 is a sparse single-point/target indicator where the mislabel is most likely to surface; the actual rate IS 1.0% residential / 0.6% commercial (NOT 0.01%), with 1.8%/1.1% 2026-2030 scenario targets, all confirmed against the workbook. Minor low-severity items: B1 unit label 'Mt CO2e' (extract) vs 'Mt CO2eq' (DB); B4 source label 'European Building Stock Observatory' (extract) vs 'demo_pjan' (context); B5a/B5b afterReport series skip 2022 (jump 2021->2023, plausible but gap). afterReport plausibility OK: B1 481.8->437.3->430.7, B2 4878.7->4213.6->4037.9, B5a 0.4558->0.42, B5b 0.3874->0.322 all continue established trends.

Flagged items:

MEDIUM **B4 (dwellings)** — [index_scale_ambiguity] Values stored as ratio (1.0=2005 base) but unit label is '% of 2005'. If renderer treats values literally with the unit, 1.086 displays as '1.086% of 2005' instead of '108.6%'. Coherent ONLY if renderer multiplies by 100 (as it does for '%' fossil shares).
MEDIUM **B4 (floor area)** — [index_scale_ambiguity] Same '% of 2005' index-vs-percent ambiguity as B4 dwellings. Values ~1.0, unit '% of 2005'.
MEDIUM **B4 (surface residential)** — [index_scale_ambiguity] Same '% of 2005' index-vs-percent ambiguity. Values ~1.0.
MEDIUM **B4 (population)** — [index_scale_ambiguity] Same '% of 2005' index-vs-percent ambiguity. Values ~1.0.
MEDIUM **B4 (surface tertiary)** — [index_scale_ambiguity] Same '% of 2005' index-vs-percent ambiguity. Values ~1.0.
HIGH **B3 (residential)** (year 2020) — [fraction_vs_percent] Value 0.010 stored with unit '%'. Workbook value is a FRACTION of building stock (0.01 = 1.0% per year), NOT 0.01%. Description text correctly reads '1.0%'. A renderer doing value+'%' shows '0.01%' (100x too small). Coherent ONLY if renderer multiplies fraction by 100 (the chapter convention used for B5a/B5b). Both data (0.010) and target (0.018) use the same fraction convention, so they are mutually consistent.
HIGH **B3 (commercial)** (year 2020) — [fraction_vs_percent] Value 0.006 stored with unit '%'. Workbook value is a FRACTION (0.006 = 0.6% per year), NOT 0.006%. Description correctly reads '0.6%'. value+'%' renderer shows '0.006%' (100x too small). Coherent only with x100 convention; data (0.006) and target (0.011) mutually consistent.

7 · Ch8

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
A1	esabcc-a1-agri-nonco2	✓	✓	5.68e-14	1.55e-14	clean
A3 (use)	esabcc-a3-fertiliser-use	✓	✓	0	0	clean
A3 (NUE)	esabcc-a3-nue	✓	✓	0	0	clean
A7	esabcc-a7-bioenergy-feedstock	✓	✓	0	0	clean
A2 (bovine, GHG)	esabcc-a2-bovine-ghg	✓	—	0.0493	0.0475	clean
A2 (bovine, production)	esabcc-a2-bovine-production	✓	—	0	0	clean
A2 (bovine, intensity)	esabcc-a2-bovine-ghg-intensity	✓	—	0.0049	0.0317	clean
A2 (dairy, GHG)	esabcc-a2-dairy-ghg	✓	—	0.0047	0.0049	clean
A2 (dairy, production)	esabcc-a2-dairy-production	✓	—	0.048	0.032	clean
A2 (dairy, intensity)	esabcc-a2-dairy-ghg-intensity	✓	—	4.9e-05	0.0072	clean
A2 (pig, GHG)	esabcc-a2-pig-ghg	✓	—	0.005	0.0174	clean
A2 (pig, production)	esabcc-a2-pig-production	✓	—	0.004	0.018	clean
A2 (pig, intensity)	esabcc-a2-pig-ghg-intensity	✓	—	0.00044	0.0353	clean
A4 (bovine, consumption)	esabcc-a4-bovine-consumption	✓	—	0	0	clean
A4 (bovine, production)	esabcc-a4-bovine-production	✓	—	0	0	clean
A4 (bovine, herd)	esabcc-a4-bovine-herd-size	✓	—	0.0042	0.007	clean
A4 (dairy, consumption)	esabcc-a4-dairy-consumption	✓	—	0.005	0.0062	clean
A4 (dairy, production)	esabcc-a4-dairy-production	✓	—	0.048	0.032	clean
A4 (dairy, herd)	esabcc-a4-dairy-herd-size	✓	—	0.0049	0.0228	clean
A4 (pig, consumption)	esabcc-a4-pig-consumption	✓	—	0.003	0.0159	clean
A4 (pig, production)	esabcc-a4-pig-production	✓	—	0.004	0.018	clean
A4 (pig, herd)	esabcc-a4-pig-herd-size	✓	—	0.047	0.0337	clean
A5 (bovine)	esabcc-a5-bovine-consumption	✓	—	0	0	clean
A5 (dairy)	esabcc-a5-dairy-consumption	✓	—	0.05	0.0112	clean
A5 (pig)	esabcc-a5-pig-consumption	✓	—	0.05	0.0225	clean
A6	esabcc-a6-food-waste	✓	—	0	0	clean

Notes: All 26 Ch8/Agriculture indicators CLEAN. DERIVED (A1, A3-use, A3-NUE, A7): recompute reproduces extract.json exactly (A1 worst 5.7e-14; A3-use/NUE/A7 == 0); A1 Value=sum of 4 EEA components (Other=residual), A3-use=Inorganic+Organic, A3-NUE=raw FAOSTAT % ratio (confirmed ratio/%), A7=Cereal+Oilseed. All .ts ts_points match extract within 4-sig-fig rounding (worst abs 0.05, all <0.05% rel). A2 INTENSITY-CONSISTENCY VERDICT: PASS — for all three animals (bovine/dairy/pig) the stored GHG-intensity series equals GHG-emissions/Production to 0% rel across every year (workbook formulas I11/I12, I3/I4, I19/I20); no wrong seriesPos, no stale intensity. A6 BREAKDOWN VERDICT: PASS — components sum exactly to 131 kg/cap/yr. A5 targets 48/301/187 match workbook 2050 benchmark series. A4 herd sizes plausible (pig ~140m, dairy ~21m, bovine non-dairy ~60m; bovine+dairy ~81m total cattle consistent with EU; task's 75-78m hint refers to total cattle, stored is non-dairy split). Several Fig59 unit cells are MISLABELED ('Mt CO2e' for dairy/pig consumption, 't CO2e/t' for dairy/pig herd) but .ts uses unitOverride and the VALUES are correct. TWO-WORKBOOK DIFF VERDICT: the laptop copy ('-laptop9883.xlsx') is an OLDER/divergent version with different sheet names and structure. Value diffs exist in NON-PUBLISHED auxiliary rows only: (1) A3 sheet row6 is 'Nitrogen Use Efficiency' (FAOSTAT %) in canonical vs 'Use per ha' (kg N/ha) in laptop — laptop LACKS the published NUE row entirely; (2) A1 demand-side scenario block is offset by one row but contains the same values; (3) crop-area sheet differs (feeds the laptop's use-per-ha, not a published Ch8 indicator); (4) A5 'Average 2021-2050'/'RoC' trajectory rows differ (different future assumptions). CRITICAL: every PUBLISHED indicator series is byte-identical between the two files OR sourced solely from the canonical file — A1 Total row3 diff=[], A3-use Total row3 identical, A2/A4/A6/A7 sheets 0 numeric diffs. No published Ch8 indicator value is at risk from the divergence. Recommendation: the stale laptop copy should be removed to avoid future divergent-source confusion, but it does not affect current deliverables.

8 · Ch9

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
L1	esabcc-l1-lulucf-net	✓	✓	0.4865	0.153	clean
L3	esabcc-l3-afforestation	✓	✓	0	0	clean
L4	esabcc-l4-deforestation	✓	✓	0	0	clean
L5	esabcc-l5-settlement-area	✓	✓	0	0	clean
L6	esabcc-l6-forest-sink	✓	✓	0	0	clean
L7	esabcc-l7-nonforest-lulucf	✓	✓	0	0	clean
L8	esabcc-l8-bioenergy-use	✓	✓	0	0	clean
L2 (cropland)	esabcc-l2-cropland-area	✓	—	0	0	clean
L2 (grassland)	esabcc-l2-grassland-area	✓	—	0	0	clean
L2 (wetland)	esabcc-l2-wetland-area	✓	—	0	0	clean
L2 (settlements)	esabcc-l2-settlements-area	✓	—	0	0	clean
L2 (forest)	esabcc-l2-forest-area	✓	—	0	0	clean
L2 (other)	esabcc-l2-other-area	✓	—	0	0	clean

Notes: Ch9/LULUCF audit: ALL 13 indicators CLEAN. DERIVED (L1,L3,L4,L5,L6,L7,L8): raw-data integrity perfect (ts_points == extract.json for all non-afterReport years; only L1 afterReport 2023/2024 absent from extract, as expected). Derivation recompute vs extract.json: L3,L4,L5,L6,L7,L8 = 0.0000 worst delta; L1 worst 0.4865 Mt @2016 (0.153% rel, within tol=1.0, matches known build_all residual). SIGN CONVENTIONS all correct: L1 net negative (sink, -342->-244 reported, weakening); L4 deforestation negative vs L3 afforestation positive (verified contrast); L6 living-biomass sink stored positive (removal magnitude); L7 non-forest stored positive (net emissions); decomposition -L6+L7 + forest-non-living-pools reconciles to L1. Unit conversions verified: L1/L7 kt->Mt /1000, L6 kt C->Mt CO2 x3.664/1000, L8 GWh->TWh /1000. L1 afterReport 2023=-215.5 (series-weakest) / 2024=-231.0 plausible, no anomalous jumps. L2 TRANSCRIPTION VERDICT: PERFECT. All 6 series x 17 years (102 hand-entered values in build.py _L2) match the chapter workbook 'L2. Areas per land-use category' (rows 3-8, Mha) to 4 decimal places with ZERO deviation. LAND-AREA SUM STABILITY: 6 categories sum to 423.539-423.541 Mha across all 17 years, total drift only 0.0013 Mha -> no transcription error in any series. PER-CATEGORY 2005->2021 CHANGE (last-first of stored series): cropland -6.528, grassland -0.325, wetland +0.249, settlements +3.269, forest +3.472, other -0.137 Mha -- these reproduce the prompt's expected magnitudes (cropland ~-6.5, forest ~+3.5, settlements ~+3.3) and exactly reproduce the workbook's column-E percentage change (which itself = (last-first)/first of the area rows). ONE LOW-SEVERITY NOTE (not a DB error): the workbook's hardcoded absolute change-bar literals in cells D11-D16 (e.g. cropland -5.929, forest +3.114, settlements +3.062) -- the values actually rendered in published Figure 66 -- do NOT equal last-first of the area rows above them (diffs 0.02-0.60 Mha); they appear to be an older data vintage, while the column-E % and the stored DB area path use the newer area series. The database faithfully stores the (authoritative) area time series; extract.json Figure 66 carries no underlying data (n=0 per series), confirming why L2 required hand transcription. No frozen series, no year offsets, no unit mismatches detected anywhere in Ch9.

9 · Finance/Innovation (F1-F5) + cross-cutting

Code	Indicator id	Raw OK	Deriv OK	worst Δabs	worst %	Flags
F1	esabcc-f-fossil-subsidies	✓	—			clean
F2	esabcc-f-green-bonds	✓	—			clean
F2 (share)	esabcc-f-green-bonds-share	✓	—			high
F3	esabcc-f-gerd	✓	—			clean
F4	esabcc-f-climate-patents-share	✓	—			clean
F5	esabcc-f-cleantech-investment	✓	—			high

Notes: Raw-data integrity for all 51 label-matched + 34 position-matched series confirmed: committed esabcc-indicators.ts reproduces the report's extract.json to 4-significant-figure rounding for every non-afterReport point (single benign outlier: I2 steel trade 2009=0.143, a genuine financial-crisis crossing). The dominant issue is the percent-encoding inconsistency, which is a display/scale defect rather than a wrong underlying value.

Flagged items:

HIGH **F2 (share)** — [unit] Stored as FRACTION (0.0008..0.0808) with unit '%'. Renderer shows value+unit with no x100, so this displays as '0.0674 %' for 2022 instead of the intended 6.74%. Part of the database-wide percent-encoding inconsistency (see cross-cutting finding).
HIGH **F5** — [unit] Stored as FRACTION (0.0002..0.0007) with unit '% of GDP'. EU clean-tech investment ~0.07% of GDP (2021) is stored as 0.0007, so it displays as '0.0007 % of GDP' (100x too small) and is 100x out of scale versus F3 GERD (2.31) which shares the unit '% of GDP'.

10 · Methodology & data pipeline

Pipeline. The ESABCC report ('Towards EU climate neutrality', 2024) ships a consolidated underlying-data workbook (one sheet per Figure). *scripts/esabcc-indicators/extract.json* is the machine extract of those sheets and is treated as the authoritative published value. *scripts/esabcc-indicators/build.py* generates *esabcc-indicators.ts* from it (rounding to 4 significant figures), and post-report 'afterReport' points are layered on by the live-source refresh. Separately, the seven chapter working workbooks (Ch3–Ch9) hold the detailed derivations; *scripts/esabcc-indicator-derivations/chapters/chN.py* reverse-engineer each published figure from the deepest raw rows so it recomputes from source.

Checks run. (a) Regenerated esabcc-indicators.ts from build.py and diffed every data point against the committed file, separating afterReport points (result: identical to 4 sig figs, 1 benign outlier). (b) Confirmed the 34 position-matched (seriesPos) new indicators map to the intended Figure series. (c) Ran build_all.py: 45/45 derivations reproduce their overview figure within 1% (E1 0.06%, L1 0.15%, rest near-exact). (d) Seven parallel agents opened the chapter workbooks and verified raw inputs, recomputed derivations at tight tolerance, checked unit/sign conventions, and verified the hand-entered dataPoints (A6 food-waste split, L2 land areas, I7 project counts, B3 renovation rates) cell-by-cell against the workbooks.

What 'no error allowed' means here. Every report-derived number was checked against the report's own data; the database is a faithful copy. The findings above concern internal consistency of units/scales and the completeness of a few new series — i.e. how the data is labelled and displayed, not whether the source numbers were copied correctly.